National Institute of Open Schooling (NIOS)
Senior Secondary Course
Lesson – 3: Trigonometry Function-I
Worksheet-03

1. Draw any one of the trigonometric functions on the graph paper. Write your observations on the following:
   (i) Maximum and minimum value of the function.
   (ii) Increasing and Decreasing of the function.
2. The minute hand of a clock is 14 cm. How far does its extremity more in 25 minutes?
3. The angle of a triangle is in Arithmetic Progression and the greatest angle is doubled the least. Express the angle of triangle in degrees and radians.
4. A wheel makes 360 revolutions in half minute. Through how many radians does it turn in one second?
5. A horse is tied to a post by a rope 60 meter long. If the horse moves along the circumference of a circle always keeping the rope in tight. Find how far it will have gone when the rope has traced an angle of $120^{o.}$
6. Check the equation $2\sin^2 \emptyset - \cos \emptyset + 4 = 0$, is it possible? If not reason out why?
7. If $\cos \emptyset + \sin \emptyset = \sqrt{2}\cos \emptyset$, then show that $c\cos\theta - \sin\theta = \sqrt{2}\sin\theta$
8. Prove that in any triangle ABC: (i) $cos\frac{A+B}{2} = sin\frac{C}{2}$ (ii) $tan\frac{A+B}{2} = Cot\frac{C}{2}$
9. Draw a graph of a trigonometry function y = tanθ, from the graph check symmetric about axes and continuity of the function.
10. Prepare a table of trigonometry functions of tanθ, cotθ, sec θ and cosec θ , When θ takes values 0^o, 30^o, 45^o, 90^o and $180^{0.}$